Potential Criterion B Assessments: Obscure but Accessible Theorems

Purpose

These assessment ideas are designed for MYP Mathematics Criterion B: **Investigating Patterns**. Each task gives students the chance to:

- investigate examples,
- notice a surprising pattern,
- make a conjecture,
- test the conjecture,
- justify or prove why the result works.

The theorems below are less common than standard school topics, but they are still visual, accessible, and connected to MYP or DP Mathematics.

Geometry-Based Investigations

Theorem / Result	Student Investigation	Pattern Students May Discover	Possible Justification / Proof
Pick's Theorem	Draw polygons on square grid paper. Count the area, boundary lattice points, and interior lattice points.	The area seems to follow $$A = I + \frac{B}{2} - 1$$ where I is the number of interior points and B is the number of boundary points.	Decompose polygons into rectangles and triangles. Students can first prove the rule for simple shapes, then combine shapes.
Viviani's Theorem	Choose different points inside an equilateral triangle. Measure the perpendicular distances from the point to each side.	The three distances always add to the height of the triangle: $$d_1 + d_2 + d_3 = h$$	Use area. The large triangle can be split into three smaller triangles with the same base-side structure.

Theorem / Result	Student Investigation	Pattern Students May Discover	Possible Justification / Proof
Napoleon's Theorem	Construct equilateral triangles on the sides of any triangle. Connect the centres of the three new equilateral triangles.	The triangle formed by the centres is always equilateral.	Use rotations by 60°, vectors, or dynamic geometry. A full proof is challenging, but students can give strong visual justification.
Van Aubel's Theorem	Construct squares on the four sides of any quadrilateral. Connect the centres of opposite squares.	The two connecting segments are always equal in length and perpendicular.	Use rotations, vectors, or coordinate geometry. This is very visual and surprising.
Morley's Trisector Theorem	Trisect the angles of a triangle. Find the intersections of adjacent trisectors.	The three intersection points always form an equilateral triangle.	A full proof is difficult, but students can investigate using GeoGebra and justify using symmetry and angle reasoning.
Butterfly Theorem	Draw a circle with a chord. Pick the midpoint of the chord. Draw two more chords through this midpoint.	Certain opposite chord segments are equal in a surprising way.	Use similar triangles and circle theorems. Suitable for stronger students.
Power of a Point	Draw secants and tangents from a point outside a circle. Measure the segment lengths.	Products of lengths are equal: $$PA \cdot PB = PC \cdot PD$$ or $$PT^2 = PA \cdot PB$$	Use similar triangles. This connects well to circle geometry and algebraic reasoning.
Simson Line Theorem	Choose a point on the circumcircle of a triangle. Drop perpendiculars from the point to the three sides of the triangle.	The three feet of the perpendiculars lie on one straight line.	Use angle chasing. Students can investigate dynamically and justify special cases.
Nine-Point Circle	For a triangle, mark the side midpoints, altitude feet, and midpoints from each vertex to the orthocentre.	The nine points all lie on one circle.	Students can prove special cases using coordinate geometry. Full proof is more advanced.
Euler Line	For different triangles, construct the centroid, circumcentre, and orthocentre.	These three points lie on one straight line.	Coordinate geometry can be used for a carefully chosen triangle.
Fagnano's Problem	Inside an acute triangle, try to find the smallest possible triangle whose vertices lie on the sides of the original triangle.	The shortest triangle is the orthic triangle, formed by the feet of the altitudes.	Use reflection and unfolding arguments. This is challenging but beautiful.

Number and Algebra Investigations

Theorem / Result	Student Investigation	Pattern Students May Discover	Possible Justification / Proof
Wilson's Theorem	Calculate $(p-1)!$ modulo p for different prime numbers p.	If p is prime, then $$(p-1)! \equiv -1 \pmod{p}$$	Pair numbers with their modular inverses. This is best for DP or advanced MYP students.
Fermat's Little Theorem	Investigate powers modulo prime numbers.	If p is prime, then $$a^p \equiv a \pmod{p}$$ or, if a is not divisible by p, $$a^{p-1} \equiv 1 \pmod{p}$$	Use modular cycles or rearrangement of remainders.
Chinese Remainder Theorem	Solve mystery number problems using remainders. For example, find a number that leaves remainder 1 when divided by 3 and remainder 2 when divided by 5.	Certain systems of remainders have a unique solution modulo the product of the divisors.	Use tables first, then justify using modular cycles.
Zeckendorf's Theorem	Try to write whole numbers as sums of non-consecutive Fibonacci numbers.	Every positive integer can be written in exactly one way as a sum of non-consecutive Fibonacci numbers.	Students can justify existence using a greedy algorithm. Uniqueness is more challenging.
Beatty Sequences	Generate sequences such as $$\lfloor n\sqrt{2} \rfloor$$ and compare them with related irrational sequences.	Two carefully chosen sequences can partition the positive integers with no overlap and no gaps.	Use number lines and floor functions. Best as an enrichment investigation.
Catalan Numbers	Count valid bracket arrangements, non-crossing handshakes, or grid paths that do not cross a diagonal.	Students discover the sequence $$1,\ 2,\ 5,\ 14,\ 42, \ldots$$	Use diagrams, recursion, or lattice paths.
Bertrand's Ballot Theorem	Investigate election-counting paths where one candidate always stays ahead.	The probability follows a simple formula depending on the final number of votes.	Use lattice paths and reflection. Strong DP Applications and Interpretation link.
Nim-Sum Theorem	Play the game Nim with different piles of counters. Record winning and losing positions.	Winning positions depend on binary XOR.	Students can justify by analysing small cases and identifying the structure of losing positions.

Theorem / Result	Student Investigation	Pattern Students May Discover	Possible Justification / Proof
Kaprekar's Constant	Choose a 4-digit number, arrange the digits from largest to smallest and smallest to largest, then subtract. Repeat.	Many numbers eventually reach $$6174$$	This is excellent for investigation, though the proof is more about classification than elegant general proof.

Graph Theory and Discrete Mathematics Investigations

Theorem / Result	Student Investigation	Pattern Students May Discover	Possible Justification / Proof
Ramsey's Theorem: $R(3,3) = 6$	Represent six people as points. Colour each connection red or blue, for example friends and strangers.	There must always be either three mutual friends or three mutual strangers.	Use the pigeonhole principle. This proof is very accessible and impressive.
Sperner's Lemma	Subdivide a large triangle into many small triangles. Colour vertices using three colours, following boundary rules.	There must be at least one small triangle with all three colours.	Use a path-following argument. This is a strong Criterion B task because it is visual and surprising.
Euler's Formula for Planar Graphs	Draw connected planar networks. Count vertices, edges, and faces.	Students discover $$V - E + F = 2$$	Prove by building the graph step by step or removing edges while preserving the relationship.
Hall's Marriage Theorem	Model matching problems, such as students choosing activities or workers choosing jobs.	A perfect matching exists only when every group has enough combined options.	Use bipartite graphs and test examples. Full proof is challenging but accessible in small cases.
Cayley's Formula	Count the number of different trees that can be formed with n labelled vertices.	The number of labelled trees is $$n^{n-2}$$	Students can explore small values. A full proof using Prüfer codes is suitable for advanced students.
Sylvester–Gallai Theorem	Place points in the plane and draw lines through pairs of points.	Unless all points are on one line, there is always a line that passes through exactly two of the points.	Students can investigate many examples. Full proof is subtle but thought-provoking.
Pick-up-Sticks Planar Graph Rule	Draw or drop sticks so that they intersect. Count the number of regions created.	The number of regions can be explained using vertices, edges, and faces.	Use Euler's formula for planar graphs. This connects a physical experiment to graph theory.

Strongest Choices for MYP Criterion B

1. Pick's Theorem

Assessment title:

Can area be found by counting points?

Students investigate polygons drawn on square grid paper. They collect data for:

- the area of the polygon,
- the number of lattice points inside the polygon,
- the number of lattice points on the boundary.

They then try to discover and justify the rule:

$$A = I + \frac{B}{2} - 1$$

This is one of the strongest options because it is visual, accessible, and surprising.

2. Viviani's Theorem

Assessment title:

What stays constant inside an equilateral triangle?

Students choose points inside an equilateral triangle and measure the perpendicular distances to each side.

They discover:

$$d_1 + d_2 + d_3 = h$$

where h is the height of the triangle.

This is excellent because the proof can be based on area, which students can understand.

3. Ramsey's Theorem

Assessment title:

Can randomness force a pattern?

Students investigate connections between six people, colouring each connection in one of two colours.

They discover that there must always be either:

- three people who are all connected in the first colour, or
- three people who are all connected in the second colour.

This is a powerful task because the proof uses only the pigeonhole principle.

4. Sperner's Lemma

Assessment title:

Can colouring rules force a triangle with all three colours?

Students colour subdivided triangles according to boundary rules.

They discover that there must be at least one small triangle containing all three colours.

This is obscure, visual, and mathematically deep.

5. Van Aubel's Theorem

Assessment title:

What happens when squares are built on any quadrilateral?

Students construct squares on the sides of different quadrilaterals. They connect the centres of opposite squares.

They discover that the two connecting segments are:

- equal in length,
- perpendicular.

This is a beautiful geometry investigation for stronger students.

Recommended Assessment Structure

1. **Generate examples:** Students create diagrams, tables, or cases.
2. **Organize data:** Students record measurements or counts clearly.
3. **Identify patterns:** Students describe what changes and what stays the same.
4. **Make a conjecture:** Students write a general rule in words, symbols, or both.
5. **Test the conjecture:** Students check whether the rule works for new examples.
6. **Justify or prove:** Students explain why the rule should always work.
7. **Reflect:** Students discuss limitations, special cases, or possible extensions.

Best Overall Recommendation

The best overall assessment is:

$$\boxed{\text{Pick's Theorem: Can area be found by counting points?}}$$

It is obscure enough that most students will not already know it, but accessible enough that they can investigate it independently.

It also allows clear differentiation:

- **Standard level:** investigate rectangles, right triangles, and simple polygons.
- **Higher level:** investigate concave polygons.
- **Extension:** investigate polygons with holes and adapt the formula.